



Custom Instruction

Sapphire Soc

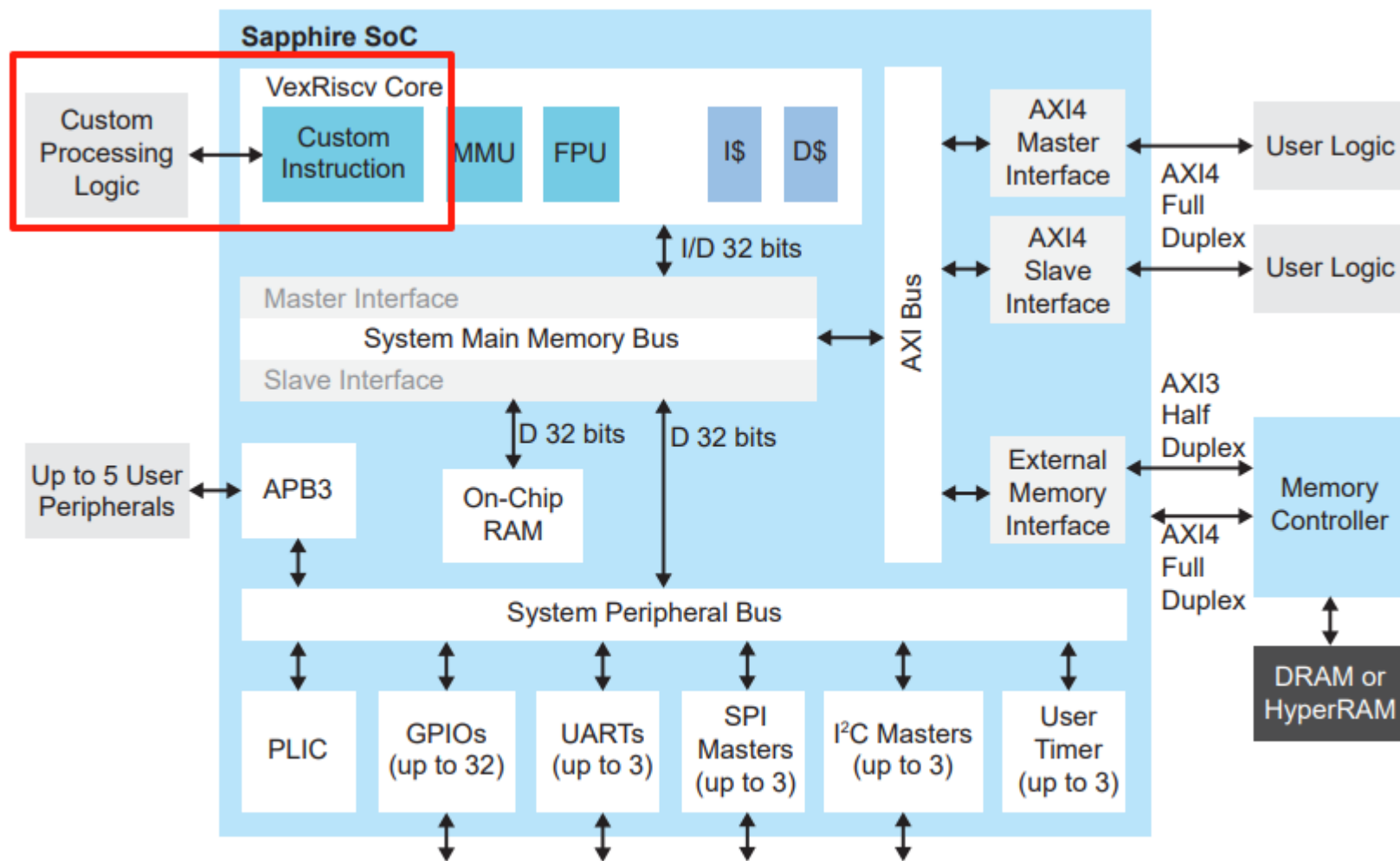
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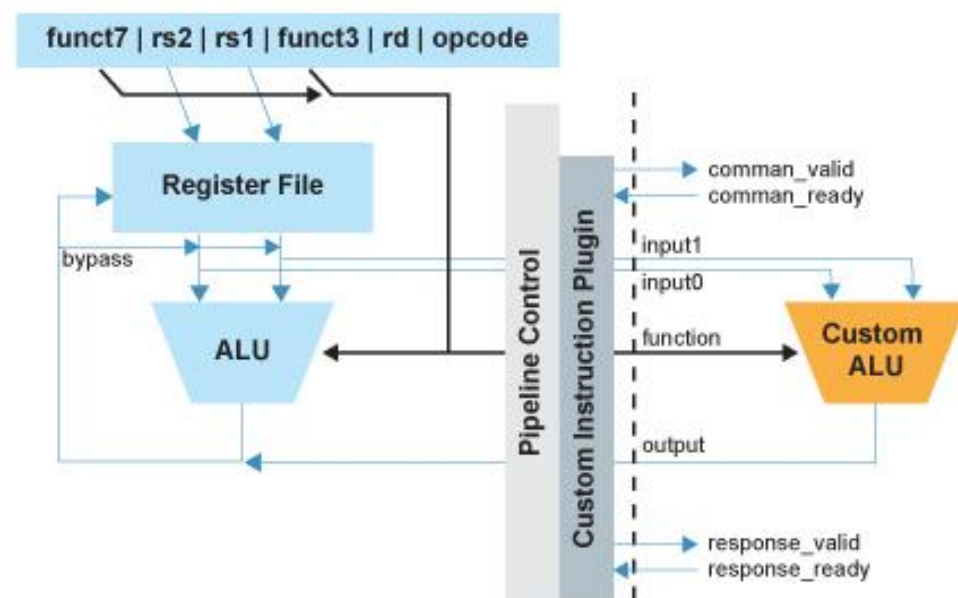
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Custom Instruction 介绍



基于SapphireSoc的
RISC-V开发
完整的视频处理流程

- 自定义ALU独立于传统ALU
- 寄存器映射
- 流水执行



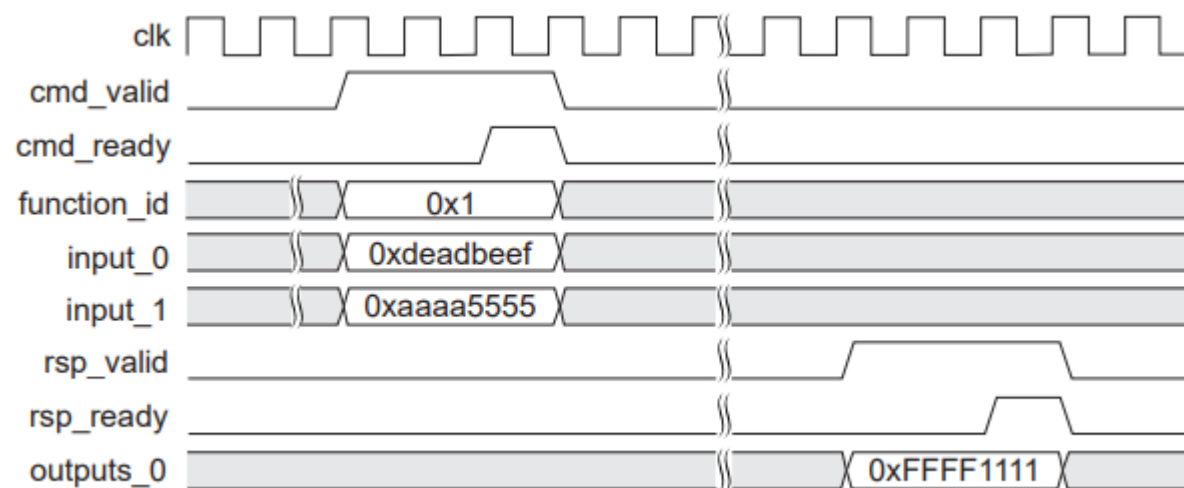
- 上图为RTL接口
- 下图为RISCV C代码接口

Port	Direction	Description
cpun_customInstruction_cmd_valid	Output	Indicates that registers rs1 and rs2 are present and ready for processing.
cpun_customInstruction_cmd_ready	Input	Indicates that the custom processing logic is ready to process register rs1 and rs2 from the CPU.
cpun_customInstruction_function_id[9:0]	Output	Function id for the custom instruction.
cpun_customInstruction_inputs_0[31:0]	Output	Register rs1 for the custom instruction.
cpun_customInstruction_inputs_1[31:0]	Output	Register rs2 for the custom instruction.
cpun_customInstruction_rsp_valid	Input	Indicates that the custom instruction result is available.
cpun_customInstruction_rsp_ready	Output	Indicates that the CPU is ready to accept the custom instruction result.
cpun_customInstruction_outputs_0[31:0]	Input	Result of the custom instruction.

opcode_R()

Usage	opcode_R(opcode, func3, func7, rs1, rs2)
Include	driver/riscv.h
Description	Define an opcode for the custom instruction.
Example	<pre>#define tea_1(rs1, rs2); opcode_R(CUSTOM0, 0x00, 0x00, rs1, rs2);</pre>

- 右图为RTL接口握手时序图
- 通过valid/ready信号实现命令/结果传输握手



Custom Instruction 示例

```
1 module simple_alu_instruction (
2     input          clk,
3     input          reset,
4     input          cmd_valid,
5     output         cmd_ready,
6     input[9:0]     cmd_function_id,
7     input[31:0]    cmd_inputs_0,
8     input[31:0]    cmd_inputs_1,
9     output         rsp_valid,
10    input          rsp_ready,
11    output[31:0]    rsp_outputs_0
12 );
13
14
15 reg [31:0]        alu_result;
16 reg              alu_result_ready;
17 reg              alu_busy;
18
19
20
21 always@(posedge clk or posedge reset) begin
22     if (reset) begin
23         alu_result        <= 32'd0;
24         alu_result_ready  <= 1'b0;
25         alu_busy          <= 1'b0;
26     end else begin
27         if (cmd_valid & cmd_ready) begin
28             case (cmd_function_id[1:0])
29                 2'd0:
30                     begin
31                         //user function code 0 here to set alu_result
32                     end
33             endcase
34         end
35     end
36 end
37
38
39 assign rsp_outputs_0    = alu_result;
40 assign rsp_valid        = rsp_ready & alu_result_ready;
41 assign cmd_ready        = rsp_ready & (!alu_result_ready) & (!alu_busy) & (!rsp_valid);
42
43 endmodule
```

➤ 左图为一个CI的简单模板：

- cmd命令组进行命令和rs1/rs2输入
- 通过10位func id判断并实现不同的功能运算
- rsp命令组进行计算结果输出

```
14 void software_tinyEncrypt (uint32_t v0, uint32_t v1, uint32_t *rv0, uint32_t *rv1) {
15     uint32_t sum=0, i;
16
17     uint32_t delta=0x9e3779b9;
18
19     uint32_t k0=0x01234567,
20             k1=0x89abcdef,
21             k2=0x13579248,
22             k3=0x248a0135;
23
24     for (i=0; i < 1024; i++) {
25         sum += delta;
26         v0 += ((v1<<4) + k0) ^ (v1 + sum) ^ ((v1>>5) + k1);
27         v1 += ((v0<<4) + k2) ^ (v0 + sum) ^ ((v0>>5) + k3);
28
29     }
30     *rv0=v0;
31     *rv1=v1;
32 }
```

- Example 中的 Tea算法 CI Demo
- 上图为软件实现
- 下图为CI的软件调用接口

```
1 #define tinyEncryption_lowerword(rsl, rs2) opcode_R(CUSTOM0, 0x00, 0x00, rsl, rs2)
2 #define tinyEncryption_upperword(rsl, rs2) opcode_R(CUSTOM0, 0x01, 0x00, rsl, rs2)
3
```

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